

Theoretical Condensed Matter: Stage 1 — Physical Foundations, Scales, and Effective Descriptions

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ABSTRACT: This eight-week Stage establishes the physical and mathematical foundation needed to begin a systematic study of theoretical condensed matter physics. Standard undergraduate quantum mechanics, classical mechanics, electromagnetism, thermodynamics, statistical mechanics, multivariable calculus, elementary linear algebra, and ordinary differential equations are assumed to have been encountered previously, but the pieces required downstream are reconstructed explicitly rather than treated as hidden prerequisites. No prior solid-state physics, second quantization, many-body Green functions, field theory, renormalization group, superconductivity, or band topology is assumed. The Stage develops operational quantum mechanics and controlled approximation methods, equilibrium ensembles and quantum statistics, gauge coupling and the microscopic electron–nuclear Hamiltonian, Born–Oppenheimer scale separation, collective low-energy descriptions, and projected effective Hamiltonians. Its purpose is to make the transition from microscopic constituents to controlled low-energy degrees of freedom explicit before crystal structure, reciprocal space, phonons, and Bloch electrons are constructed in Stage 2. The core workload is approximately 75–90 focused hours, depending on the amount of prerequisite reconstruction required.

Contents

1	Stage 1: Physical Foundations, Scales, and Effective Descriptions	2
1.1	Week 1 — Operational quantum mechanics and mathematical language	3
1.2	Week 2 — Controlled approximation and identical particles	4
1.3	Week 3 — Ensembles, density operators, and thermodynamic potentials	5
1.4	Week 4 — Quantum statistics, density of states, and the thermodynamic limit	6
1.5	Week 5 — Gauge structure, microscopic matter, and natural scales	7
1.6	Week 6 — Born–Oppenheimer separation and adiabatic structure	8
1.7	Week 7 — Collective variables, long-wavelength limits, and emergence	9
1.8	Week 8 — Low-energy projection and effective Hamiltonians	10
2	References and online resources	12
2.1	Quantum mechanics and mathematical tools	12
2.2	Statistical mechanics	12
2.3	Microscopic matter, scales, and collective descriptions	13
2.4	Effective descriptions	13

1 Stage 1: Physical Foundations, Scales, and Effective Descriptions

Stage 1 reconstructs the quantum-mechanical and statistical tools that will be used throughout the program, then applies them to microscopic matter, separation of scales, collective descriptions, and controlled low-energy projection. Each week closes a dependency needed by the next; later condensed-matter structures are introduced only when their prerequisites have been established.

1.1 Week 1 — Operational quantum mechanics and mathematical language

Step	Action	Exact task and source
1	CHECK	Before the assigned reading, verify that you can diagonalize a generic 2×2 Hermitian matrix, evaluate a basic commutator such as $[x, p^2]$, normalize a Gaussian and state a Fourier convention, recover the harmonic-oscillator spectrum from ladder operators, and differentiate a simple partition function to obtain $\langle E \rangle$ and C_V . Repair quantum-mechanics or mathematical gaps during Weeks 1–2; record statistical-mechanics gaps for explicit reconstruction in Weeks 3–4 rather than treating them as hidden prerequisites.
2	READ	D. Tong, <i>Quantum Mechanics</i> , Ch. 3, Secs. 3.1–3.4 only: states, inner products, observables, Hermitian operators, momentum eigenstates/representation, measurement, expectation values, and commutators. Stop before Sec. 3.5. [PDF]
3	REFRESH	Use Tong Ch. 2, Sec. 2.2 only if the harmonic-oscillator ladder construction is not already fluent. [PDF]
4	READ	Sakurai–Napolitano, <i>Modern Quantum Mechanics</i> , 3rd ed., Sec. 3.1.4 and Secs. 3.2.1, 3.2.4–3.2.5 only: angular-momentum algebra, spin-1/2, Pauli two-component formalism, and spin rotations. Skip the experimental interludes. The goal is the minimal spin language used in Weeks 4 and later Stages.
5	DERIVE	For a continuous unitary transformation $U(\epsilon) = e^{-i\epsilon G/\hbar}$, expand $U^\dagger H U$ to first order and derive the infinitesimal symmetry condition $[H, G] = 0$. Derive the Heisenberg equation $dO/dt = (i/\hbar)[H, O] + \partial_t O$ and conclude that G is conserved only when both $[H, G] = 0$ and $\partial_t G = 0$.
6	DERIVE	Fix one Fourier convention, derive $\delta^{(d)}(\mathbf{r}) = \int d^d k e^{i\mathbf{k} \cdot \mathbf{r}} / (2\pi)^d$, and place the system in a rectangular box with periodic boundary conditions, $k_i = 2\pi n_i / L_i$. From the k -space cell volume derive $\sum_{\mathbf{k}} \rightarrow V \int d^d k / (2\pi)^d$ in the large-volume limit.
7	SOLVE	Four short original checks: (i) for a specified 2×2 Hamiltonian and generator, determine whether the transformation is a symmetry and whether the generator is conserved; (ii) transform one operator and state between two orthonormal bases and verify completeness; (iii) exponentiate a Pauli-matrix generator and verify the corresponding spin rotation; (iv) compute selected harmonic-oscillator matrix elements and verify the adopted Fourier normalization on a Gaussian.
8	OUTCOME	Use state/operator/basis/spin/Fourier language without hidden convention changes, distinguish Hamiltonian invariance from conservation, and state the explicit-time-dependence condition required for a symmetry generator to define a conserved observable.

1.2 Week 2 — Controlled approximation and identical particles

Step	Action	Exact task and source
1	READ	MIT 8.06 (Zwiebach), <i>Quantum Physics III</i> , Spring 2018, Ch. 1: non-degenerate and degenerate time-independent perturbation theory. Read through the construction and validity discussion; omit later fine-structure applications. [PDF]
2	READ	D. Tong, <i>Topics in Quantum Mechanics</i> , Sec. 2.1, “The Variational Method,” only. [PDF]
3	READ	MIT 8.06, Ch. 8 <i>Identical Particles</i> , Secs. 8.1–8.4: exchange degeneracy, permutation operators, symmetrizers/antisymmetrizers, Slater determinants, and occupation-number labeling. Stop before Sec. 8.5. [PDF]
4	DERIVE	Derive $E_n^{(1)}$, $ n^{(1)}\rangle$, and $E_n^{(2)}$ for nondegenerate perturbation theory. Identify the actual expansion parameters as matrix-element-to-gap ratios, schematically $ \lambda V_{mn}/(E_n^{(0)} - E_m^{(0)}) $, and show why near-degeneracy invalidates the nondegenerate organization and requires diagonalizing PVP in the degenerate subspace.
5	DERIVE	Construct the two-particle space $\mathcal{H}_1 \otimes \mathcal{H}_2$ and operators $A_1 = A \otimes I$, $A_2 = I \otimes A$. From the permutation operator build the symmetrizer and antisymmetrizer, prove the Rayleigh–Ritz upper bound, and construct normalized bosonic states and fermionic Slater determinants. Show why repeated one-particle orbitals annihilate a fermionic determinant and derive the occupation restrictions $n_i \in \{0, 1\}$ for fermions and $n_i = 0, 1, 2, \dots$ for bosons. Take the assignment of spin-1/2 electrons to fermionic statistics as an established spin–statistics result; its relativistic proof is not part of this Stage.
6	SOLVE	Tong <i>Topics in QM</i> , Example Sheet 2, Problem 1(a–b), on variational estimates for the quartic oscillator. Then solve MIT 8.06 Problem Set 1, Problem 1(b), comparing exact and perturbative eigenvalues of a 2×2 Hamiltonian, and MIT 8.06 Problem Set 9, Problem 1(a) plus Problem 3(a–b), on exchange symmetry and three identical fermions/bosons. [PROBLEMS] [PROBLEMS] [PROBLEMS]
7	DEEPEN	Sakurai–Napolitano, <i>Modern Quantum Mechanics</i> , 3rd ed., Secs. 5.1–5.2 (nondegenerate/degenerate time-independent perturbation theory), 5.4 (variational methods), and 7.5 (multiparticle states and Slater determinants), only as a canonical reference if a derivation above is not secure.
8	OUTCOME	Given $H = H_0 + \lambda V$, diagnose the relevant matrix-element/gap ratio, choose nondegenerate versus degenerate perturbation theory correctly, and construct legal symmetric or antisymmetric many-particle states with the correct occupation restrictions.

1.3 Week 3 — Ensembles, density operators, and thermodynamic potentials

Step	Action	Exact task and source
1	READ	D. Tong, <i>Statistical Physics</i> , Ch. 1, Secs. 1.2–1.4.3: microcanonical, canonical, chemical potential, grand canonical ensemble, and extensive/intensive variables. [PDF]
2	READ	Sakurai–Napolitano, <i>Modern Quantum Mechanics</i> , 3rd ed., Secs. 3.4.2–3.4.5 only: ensemble averages, the density operator, its time evolution/continuum form, and the bridge to quantum statistical mechanics. The role is distinct from Tong: connect ensemble probabilities to the operator ρ .
3	DERIVE	From an ensemble $\{w_i, \psi_i\rangle\}$ construct $\rho = \sum_i w_i \psi_i\rangle\langle\psi_i $ and establish $\rho^\dagger = \rho$, $\rho \geq 0$, $\text{Tr } \rho = 1$, and $\langle A \rangle = \text{Tr}(\rho A)$. Show that $\text{Tr } \rho^2 = 1$ for a pure state and $\text{Tr } \rho^2 < 1$ for a genuinely mixed state.
4	DERIVE	Starting from a weakly coupled reservoir, derive $p_n \propto e^{-\beta E_n}$ and then $p_n \propto e^{-\beta(E_n - \mu N_n)}$ when particle number is an exchanged conserved quantity, $[H, N] = 0$. Construct $\rho_{\text{can}} = e^{-\beta H}/Z$ and $\rho_{\text{gc}} = e^{-\beta(H - \mu N)}/\Xi$, and derive $F = -k_B T \ln Z$ and $\Omega = -k_B T \ln \Xi$.
5	DERIVE	Derive $\langle E \rangle = -\partial_\beta \ln Z$, $\Delta E^2 = k_B T^2 C_V$, $\langle N \rangle = \beta^{-1} \partial_\mu \ln \Xi$, and the corresponding particle-number fluctuation relation.
6	SOLVE	Tong Statistical Physics Sheet 1, Problems 2(ii–iii), 7, and 8. These respectively test fluctuation/entropy identities, maximum-entropy derivations of all three ensembles, and grand-canonical number fluctuations. [PROBLEMS]
7	OUTCOME	Derive the ensemble appropriate to specified constraints, construct its density operator, distinguish pure from mixed statistical states, and obtain thermodynamic observables and fluctuations directly from Z , Ξ , or ρ rather than quoting memorized formulas.

1.4 Week 4 — Quantum statistics, density of states, and the thermodynamic limit

Step	Action	Exact task and source
1	READ	Tong, <i>Statistical Physics</i> , Ch. 3: Sec. 3.1 (density of states), the opening of Sec. 3.5 and Secs. 3.5.1–3.5.2 (identical quantum particles, Bose distribution, and classical limit), and Secs. 3.6–3.6.4 through the low- T Fermi gas/Sommerfeld expansion. Skip Bose condensation in Sec. 3.5.3 and stop before the astrophysical/magnetic applications that follow the Fermi-gas core. [PDF]
2	DERIVE	For $\epsilon_k = \hbar^2 k^2 / 2m$, define $g_0(E)$ as the density of one-particle states per internal state and derive it in general d , specializing to $d = 1, 2, 3$. Define $g(E) = g_s g_0(E)$ when an internal degeneracy is included, with $g_s = 2$ for spin-1/2 electrons. Then derive Bose–Einstein and Fermi–Dirac occupations from independent-mode grand partition functions.
3	DERIVE	Using the same DOS convention throughout, derive for a three-dimensional ideal electron gas at $T = 0$ the relations for k_F , E_F , $g(E_F)$, and $E_0/N = 3E_F/5$. Then reproduce the leading Sommerfeld corrections to $\mu(T)$ and $C_V \propto T$.
4	DERIVE	Show explicitly how $\Delta E / \langle E \rangle$ and $\Delta N / \langle N \rangle$ scale as $N^{-1/2}$ for ordinary additive short-range systems with finite intensive response functions, away from singular critical regimes. Explain how concentration of the relevant extensive variables supports equivalence of bulk thermodynamic observables among ensembles under these regularity/additivity assumptions; do not promote this argument to unrestricted equality of finite-size fluctuations or to systems with nonadditive thermodynamics.
5	SOLVE	Tong Statistical Physics Sheet 2, Problem 8 (DOS in $d = 1, 2$), and Sheet 3, Problems 5 and 8 (grand-canonical fermion occupations and high-/low- T Fermi gas). [PROBLEMS] [PROBLEMS]
6	OUTCOME	Given a dispersion and dimensionality, compute the DOS with an explicit degeneracy convention, obtain the equilibrium occupation physics and Fermi scale, and state precisely when thermodynamic-limit concentration supports equivalence of bulk ensemble predictions.

1.5 Week 5 — Gauge structure, microscopic matter, and natural scales

Step	Action	Exact task and source
1	READ	D. Tong, <i>Solid State Physics</i> , Ch. 1, Secs. 1.1–1.1.2 only: electromagnetic potentials, the charged-particle Hamiltonian, and gauge transformations. Stop before Sec. 1.2 (Landau levels). [PDF]
2	READ	M. L. Cohen and S. G. Louie, <i>Fundamentals of Condensed Matter Physics</i> (Cambridge University Press, 2016), Ch. 1 Sec. 1.2 and Ch. 2 Sec. 2.1 only: interacting-atom/core–valence picture and the general electron–core/valence solid Hamiltonian. Defer mean field and periodic-potential material to later Stages.
3	DERIVE	From $L = \frac{1}{2}m\dot{\mathbf{r}}^2 + q\dot{\mathbf{r}} \cdot \mathbf{A} - q\phi$, define the canonical momentum $\mathbf{p} = \partial L / \partial \dot{\mathbf{r}}$ and perform the Legendre transform $H = \mathbf{p} \cdot \dot{\mathbf{r}} - L$ to obtain $H = (\mathbf{p} - q\mathbf{A})^2/2m + q\phi$. Verify gauge covariance under $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$, $\phi \rightarrow \phi - \partial_t\chi$, and $\psi \rightarrow e^{iq\chi/\hbar}\psi$.
4	DERIVE	Fix $e > 0$, electron charge $q_e = -e$, and nuclear charge $q_I = Z_I e$. Write the full nonrelativistic Coulomb Hamiltonian $T_N + T_e + V_{NN} + V_{ee} + V_{eN}$ with correct signs, pair-counting factors, masses, and SI normalization. Identify only the reorganizations used in this Stage: core–valence grouping, fixed-nuclei/adiabatic separation, and later low-energy projection; mean-field theory is deferred.
5	DERIVE	Use $E(a) \sim \hbar^2/(2m_e a^2) - e^2/(4\pi\epsilon_0 a)$ to obtain the Bohr scale $a_0 \simeq 0.529 \text{ \AA}$, the Coulomb/Hartree scale $E_H \simeq 27.2 \text{ eV}$ and hydrogenic binding scale $E_H/2 \simeq 13.6 \text{ eV}$, together with $\hbar/(1 \text{ eV}) \simeq 0.658 \text{ fs}$. Interpret $\text{\AA}/\text{eV}/\text{fs}$ as useful electronic condensed-matter orders of magnitude, not universal material constants, and compare them with the nuclear/electronic mass hierarchy.
6	SOLVE	Original scale-and-gauge set: (i) verify gauge covariance of the time-dependent Schrödinger equation for a charged particle; (ii) perform a dimensional/sign audit of the microscopic electron–nuclear Hamiltonian using the stated charge convention; (iii) estimate the length, energy, and time scales for one representative valence-electron system and identify which ratios could justify an approximation.
7	OUTCOME	Write the microscopic matter Hamiltonian from physical principles, distinguish canonical from kinetic momentum, and identify which dimensionless ratios and scale hierarchies can justify later approximations.

1.6 Week 6 — Born–Oppenheimer separation and adiabatic structure

Step	Action	Exact task and source
1	READ	D. Tong, <i>Topics in Quantum Mechanics</i> , Sec. 2.3.6, “The Born–Oppenheimer Approximation,” only. Read the full derivation in that subsection; do not import the later molecular material as a new topic. [PDF]
2	READ	Cohen–Louie, <i>Fundamentals of Condensed Matter Physics</i> , Ch. 2 Sec. 2.2 only for the condensed-matter formulation of the adiabatic approximation. Stop before Sec. 2.3 (mean-field approximation).
3	DERIVE	Write $H = T_N + H_e(\mathbf{R})$ and solve $H_e(\mathbf{R}) \phi_n(\mathbf{R})\rangle = E_n(\mathbf{R}) \phi_n(\mathbf{R})\rangle$. Expand the exact state as $ \Psi\rangle = \sum_n \chi_n(\mathbf{R}) \phi_n(\mathbf{R})\rangle$ and derive the coupled nuclear equations before making an adiabatic truncation. Introduce $\mathbf{d}_{mn}^{(I)} = \langle\phi_m \nabla_I\phi_n\rangle$ and display explicitly the off-diagonal derivative couplings that mix electronic surfaces.
4	DERIVE	For a nondegenerate single-surface reduction, derive the adiabatic nuclear Hamiltonian in the form $H_n^{\text{ad}} = \sum_I \frac{[-i\hbar\nabla_I - \hbar\mathbf{A}_{nI}(\mathbf{R})]^2}{2M_I} + E_n(\mathbf{R}) + \Phi_n(\mathbf{R}),$ with $\mathbf{A}_{nI} = i\langle\phi_n \nabla_I\phi_n\rangle$ and $\Phi_n = \sum_I \frac{\hbar^2}{2M_I} \sum_{m \neq n} \langle\phi_m \nabla_I\phi_n\rangle ^2$. Distinguish this single-surface adiabatic Hamiltonian from the leading Born–Oppenheimer approximation that neglects derivative-coupling corrections and retains $E_n(\mathbf{R})$ as the potential-energy surface. Treat $\mathbf{A}_{nI}$ only as a preview; its geometric interpretation is deferred to Stage 3.
5	DERIVE	For a diatomic scaling estimate, show parametrically that electronic, vibrational, and rotational energy scales obey $E_{\text{vib}}/E_{\text{elec}} \sim (m_e/M)^{1/2}$ and $E_{\text{rot}}/E_{\text{elec}} \sim m_e/M$. Explain why the mass ratio motivates the hierarchy but is not by itself a sufficient adiabaticity criterion: electronic gaps and derivative matrix elements also control nonadiabatic mixing.
6	LECTURE	MIT 8.06, Spring 2018: L17.4 “Molecules and energy scales,” L18.1 “Born–Oppenheimer approximation: Hamiltonian and electronic states,” and L18.2 “Effective nuclear Hamiltonian. Electronic Berry connection.” L18.3 (H_2^+) is optional. [LECTURES]
7	SOLVE	Cohen–Louie, Part I, Problem I.3, on the m/M hierarchy of electronic, vibrational, and rotational molecular scales. Then perform one diagnostic transfer problem: for each of (i) a small electronic gap/avoided crossing, (ii) light nuclei, and (iii) large derivative coupling $\mathbf{d}_{mn}^{(I)}$, identify which adiabatic/BO step ceases to be controlled.
8	OUTCOME	Derive the exact electronic-basis expansion, distinguish single-surface adiabatic reduction from leading Born–Oppenheimer truncation, state the mass/gap/derivative-coupling control assumptions, and recognize nonadiabatic regimes before using a static lattice in Stage 2.

1.7 Week 7 — Collective variables, long-wavelength limits, and emergence

Step	Action	Exact task and source
1	READ	Cohen–Louie, <i>Fundamentals of Condensed Matter Physics</i> , Ch. 1 Secs. 1.3–1.8: elementary excitations, probes, dispersion relations, and interactions. The goal is the operational excitation-based description, not memorization of the catalogue of excitations.
2	READ	P. Coleman, <i>Introduction to Many-Body Physics</i> (Cambridge University Press, 2015), Introduction and Ch. 1 (“Scales and complexity”) only. Stop before Ch. 2, where the text begins the field/phonon construction reserved for later Stages.
3	DEEPEN	P. W. Anderson, “More Is Different,” <i>Science</i> 177 , 393–396 (1972). Read as an interpretive primary source after the technical work, not as a substitute for it. [PAPER]
4	DEEPEN	Optional: R. B. Laughlin and D. Pines, “The Theory of Everything,” <i>PNAS</i> 97 , 28–31 (2000), for the “quantum protectorate” viewpoint and low-energy insensitivity to microscopic detail. [PAPER]
5	DERIVE	Define $n_{\text{mic}}(\mathbf{r}) = \sum_i \delta(\mathbf{r} - \mathbf{r}_i)$ and its Fourier components. Construct $n_\ell(\mathbf{r}) = \int d^d r' K_\ell(\mathbf{r} - \mathbf{r}') n_{\text{mic}}(\mathbf{r}')$ with $\int d^d r K_\ell(\mathbf{r}) = 1$ and $\ell \gg a$, and derive $n_\ell(\mathbf{q}) = K_\ell(\mathbf{q}) n_{\text{mic}}(\mathbf{q})$. Show that a smooth kernel suppresses high- q information for $q \gtrsim \ell^{-1}$; distinguish this from the special sharp low-pass projection $\Theta(\Lambda - \mathbf{q})$.
6	DERIVE	For N equal masses M with spacing a , nearest-neighbor spring constant K , and periodic boundary conditions, diagonalize the classical monatomic harmonic chain and derive $\omega(q) = 2\sqrt{K/M} \sin(qa/2) , \quad q = \frac{2\pi n}{Na}.$ Then take the controlled long-wavelength limit $ qa \ll 1$ to obtain $\omega(q) \simeq a\sqrt{K/M} q \equiv c q $. Do not quantize the modes here; Stage 2 develops phonons completely.
7	SYNTHESIZE	Build a two-column comparison for one microscopic system: microscopic constituents/coordinates/Hamiltonian versus low-energy variables/excitations/observables. State what is derived, what is inferred, what information is discarded by the limiting operation, and what remains phenomenological.
8	OUTCOME	Explain emergence without hand-waving: identify the limiting operation or hierarchy that removes microscopic information and demonstrate at least one collective variable or long-wavelength mode arising from the microscopic description.

1.8 Week 8 — Low-energy projection and effective Hamiltonians

Step	Action	Exact task and source
1	READ	S. Bravyi, D. P. DiVincenzo and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” Sec. 1 (Introduction), especially the formulation in terms of low- and high-energy subspaces and perturbative decoupling. Use it as provenance and precision for the projection problem, not as a substitute for the derivation below. [PAPER]
2	DERIVE	Let P and $Q = 1 - P$ be orthogonal projectors, $P^2 = P = P^\dagger$, $Q^2 = Q = Q^\dagger$, and $PQ = 0$. Project $(H - E) \Psi\rangle = 0$ into the two sectors, eliminate $Q \Psi\rangle$, and derive the exact energy-dependent projected operator $H_{\text{eff}}(E) = PHP + PHQ(E - QHQ)^{-1}QHP,$ valid when $E \notin \sigma(QHQ)$.
3	DERIVE	Take $H = H_0 + \lambda V$ with P a degenerate low-energy eigenspace satisfying $PH_0P = E_0P$, $PH_0Q = 0$, and a gap $\Delta > 0$ separating E_0 from $\sigma(QH_0Q)$. Derive $H_{\text{eff}} = E_0P + \lambda PVP - \lambda^2 PVQ(QH_0Q - E_0)^{-1}QVP + O(\lambda^3),$ and identify $\lambda\ V\ /\Delta \ll 1$ as a sufficient perturbative control condition, with $\lambda\ PVQ\ /\Delta$ directly measuring low/high-sector mixing. Distinguish this approximate energy-independent Hamiltonian from the exact resolvent form. State that energy-independent effective Hamiltonians related by a unitary acting only inside P are equivalent representations, and that observables must be mapped consistently, e.g. $O_{\text{eff}} = PUOU^\dagger P$ in a unitary block-diagonalization convention.
4	SOLVE	Use the fully specified three-level benchmark $H = \begin{pmatrix} 0 & 0 & v_1 \\ 0 & 0 & v_2 \\ v_1^* & v_2^* & \Delta \end{pmatrix}, \quad v_1 , v_2 \ll \Delta.$ Derive $H_{\text{eff}}^{(2)} = -\Delta^{-1} \begin{pmatrix} v_1 ^2 & v_1 v_2^* \\ v_2 v_1^* & v_2 ^2 \end{pmatrix}$. Diagonalize the full 3×3 Hamiltonian, identify the exact dark state, and show that the nonzero low eigenvalue is $E_- = [\Delta - \sqrt{\Delta^2 + 4V^2}]/2 = -V^2/\Delta + V^4/\Delta^3 + O(V^6/\Delta^5)$ with $V^2 = v_1 ^2 + v_2 ^2$, so the second-order eigenvalue error begins at $O(V^4/\Delta^3)$.

Step	Action	Exact task and source (continued)
5	DEEPEN	Optional: relate the projected result to the unitary Schrieffer–Wolff construction $U = e^S$ and the condition that the transformed Hamiltonian be block diagonal. Do <i>not</i> carry out the Anderson-to-Kondo transformation; that is a Stage-6 application.
6	SYNTHESIZE	Return to the Week-5 microscopic matter Hamiltonian. Compare three reductions: (i) Born–Oppenheimer separation; (ii) thermodynamic/long-wavelength coarse graining; and (iii) perturbative P/Q projection. For each, state the eliminated information, control assumption, retained degrees of freedom and observables, required operator mapping when relevant, and validity domain.
7	GATE	Cold-start exit problem: given an unfamiliar microscopic Hamiltonian with an explicit hierarchy, identify the low-energy sector, justify an appropriate reduction, derive the leading effective Hamiltonian, estimate the neglected scale/order, and state which symmetry and observable content must be preserved or transformed consistently.
8	OUTCOME	Construct and audit a controlled low-energy Hamiltonian rather than treating “effective theory” as a slogan. This is the Stage-1 exit competency required before crystal structure and Bloch physics are introduced in Stage 2.

2 References and online resources

2.1 Quantum mechanics and mathematical tools

Resource	Use in the program and access
David Tong, <i>Quantum Mechanics</i>	Week 1 operational refresher on states, observables, commutators, momentum representation, Fourier language, and the harmonic oscillator when needed. [COURSE]
J. J. Sakurai and J. Napolitano, <i>Modern Quantum Mechanics</i> , 3rd ed. (Cambridge University Press, 2021)	Selective canonical reference for spin-1/2, density operators, perturbation/variational methods, and multiparticle antisymmetry in Weeks 1–3.
MIT 8.06, <i>Quantum Physics III</i> , B. Zwiebach, Spring 2018	Perturbation theory and identical particles in Week 2; molecular energy scales and the adiabatic/Born–Oppenheimer construction in Week 6. [COURSE] [LECTURES]
David Tong, <i>Topics in Quantum Mechanics</i>	Variational method in Week 2 and Born–Oppenheimer derivation in Week 6; used as a complementary derivational source rather than a second full quantum-mechanics course. [COURSE] [PROBLEMS]

2.2 Statistical mechanics

Resource	Use in the program and access
David Tong, <i>Statistical Physics</i>	Main Stage-1 statistical-mechanics source: ensembles, chemical potential, grand canonical ensemble, density of states, Bose–Einstein and Fermi–Dirac statistics, and low-temperature Fermi-gas physics in Weeks 3–4. [COURSE] Sheets 1–3: [PROBLEMS] [PROBLEMS] [PROBLEMS]

2.3 Microscopic matter, scales, and collective descriptions

Resource	Use in the program and access
David Tong, <i>Solid State Physics</i>	Week 5 minimal-coupling and gauge-structure bridge. The assignment stops before Landau-level physics, which belongs to a later Stage. [COURSE]
M. L. Cohen and S. G. Louie, <i>Fundamentals of Condensed Matter Physics</i> (Cambridge University Press, 2016)	Microscopic electron-core/valence description, Born–Oppenheimer context, the Part-I Born–Oppenheimer scaling problem, and the elementary-excitation viewpoint in Weeks 5–7. Mean-field and periodic-potential developments are deferred to later Stages.
P. Coleman, <i>Introduction to Many-Body Physics</i> (Cambridge University Press, 2015)	Week 7 synthesis of scales, complexity, and emergent low-energy description. Second quantization and the later many-body formalism are deliberately deferred.
P. W. Anderson, “More Is Different,” <i>Science</i> 177 , 393–396 (1972)	Interpretive primary source used only after the technical coarse-graining and collective-mode constructions of Week 7. [PAPER]
R. B. Laughlin and D. Pines, “The Theory of Everything,” <i>PNAS</i> 97 , 28–31 (2000)	Optional Week-7 extension on organizing principles and low-energy insensitivity to microscopic detail. [PAPER]

2.4 Effective descriptions

Resource	Use in the program and access
S. Bravyi, D. P. DiVincenzo and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” <i>Annals of Physics</i> 326 , 2793–2826 (2011)	Week 8 provenance and precision for the low-/high-energy decoupling problem. Only the introductory formulation is core here; the Anderson-to-Kondo application is reserved for Stage 6. [PAPER]